BIOMEDICAL LIMITED

ANNUAL PRODUCT QUALITY REVIEW

(YEAR – 2025)

Protocol No. : **BML/QA/APQR/QA/002**

Protocol Date : 23/04/2024

Total No. of Pages :

PRODUCT NAME : Bioflex Mannitol 10%w/v Intravenous Injection B.P 500ml

GENERIC NAME : Mannitol 10%w/v Intravenous Injection B.P 500ml

COMPOSITION : Each 500ml contains

Mannitol 50g

Water for injection B.P q.s

PACK SIZE : 500ml in a printed PVC pouch

PACKAGING TYPES AVAILABLE : 500ml Polyvinylchloride PVC pouch

Corrugated cardboard boxes carton

MASTER FORMULAR NUMBER : BML/QA/MFR/011

SHELF LIFE: 3 Years

TIME PERIOD COVERED : JANUARY – DECEMBER, 2025

PRODUCT CODE : 1M

BIOMEDICAL LIMITED

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Sr. No	Subject	Name	Designation	Sign/Date
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3	Approved by	Koroma Abigael A.	Quality Assurance Manager	
4	Authorized by	Agboola L. A	Factory Manager	

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1.0 SCOPE:

This covers internal annual products quality review, and all these parameters are monitored; changes in physical appearance of the product, chemical analysis (pH, Assay), microbiological analysis (Sterility and Pyrogen), and post market surveillance.

2.0 DESCRIPTION OF PRODUCT

Bioflex Mannitol 10%w/v Intravenous Injection is a clear colorless sterile solution in a PVC pouch with double port for easy administration. Formulated for the treatment of……, the hypertonic solution is carefully controlled to ensure the mannitol level of 50g/500ml. Careful control of all other parameters as mentioned in the scope and under the section for the review of the quality control of the product in this piece are assured at all times.

3.0 REVIEWS

3.1 REVIEW OF PRODUCT BATCHES MANUFACTURED

A batch of Bioflex Mannitol 10% w/v Intravenous Injection comprises of about 3,050 bags. Total No. of Batches produced within the period January 2025 to December 2025 is 7 batches. I.e. from Batch no.:1M25001 to 1M25007; all the batches produced have shown reliability of the manufacturing and the testing procedures in delivering constant and reproducible good product quality.

3.2 REVIEW OF STARTING AND PRIMARY PACKAGING MATERIALS USED IN THE FPP

The starting materials for the manufacture of Bioflex Mannitol 10%w/v Intravenous Injection are Mannitol (active ingredient) and Water for injection (excipient). Mannitol is sourced from approved vendors while water for injection is produced in house. The starting materials are each tested to comply with respective pharmacopeia and in-house specifications. The table below shows the different consignment of Mannitol received and used in the manufacture of Bioflex Mannitol 10%w/v Intravenous Injection within the year under review. All the batches of the active material received met the specifications of pharmacopeia.

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3.2.1 VENDORS

All the raw materials used in the production of these batches are sourced from the approved vendors only.

3.2.2 RAW MATERIAL (API) MANUFACTURER/ SUPPLIER PERFORMANCE REVIEW

Name of API	Manufacturer (Name and Address)	Supplier (Name and Address)	Consignment / Batches Received	Consignment Approved	Consignment Rejected	Causes of Rejection
Mannitol	Shijia Zhuancy xuaxu Pharmaceutical Co. Ltd, China.	Bdv Behrens, Gmbh Hamburg, Germany.	**1**	**1**	**Nil**	**Nil**

3.2.3 PRIMARY PACKAGING MATERIAL

The FPP is contained in poly vinyl chloride (PVC) self-collapsible soft bag (pouch) bought as a finished product from the manufacturer of the PVC pouch. Manufacturer COA of the PVC is reviewed and testing done in-house to confirm the quality specifications stated by the manufacturers as conform to in-house specifications set for the primary packaging material.

Name of Primary packaging Material	Manufacturer (Name and Address)	Supplier (Name and Address)	Consignments / Batches Received	Consignment Approved	Consignment Rejected	Causes of Rejection
PVC pouches	Jiangsu Medwell Medical Equipment Co. Ltd. China	Jiangsu Medwell Medical Equipment Co. Ltd. China	7	7	Nil	Nil

3.3 REVIEW AND STATISTICAL ANALYSIS OF QUALITY CONTROL AND IN-PROCESS CONTROL

3.3.1 SPECIFICATIONS

Appearance: Clear, colorless and particles free solution

Mannitol B.P.: 95 – 105%

pH: 4.50 – 7.50

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3.3.2 IN-PROCESS RESULTS:

The summary of in-process results of all the batches manufactured was prepared and the table showing the results presented. The summary of in-process results for the chemical analysis and microbiological analysis of the batches manufactured is shown.

Analysis for Inprocess Result

S/N	**BATCH NO**	**Mannitol** Assay (95 – 105%)	**pH** (sample) (4.5 – 7.5)	**pH** (WFI) (5.0 – 7.0)
1	1M25001	97.02	6.27	6.3
2	1M25002	98.2	6.3	6.36
3	1M25003	97.84	6.28	6.4
4	1M25004	98.69	6.08	6.42
5	1M25005	96.07	5.95	6.25
6	1M25006	97.05	6	6.3
7	1M25007	97.77	6.03	6.25

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Chemical/Microbial Analysis for Finished product Result

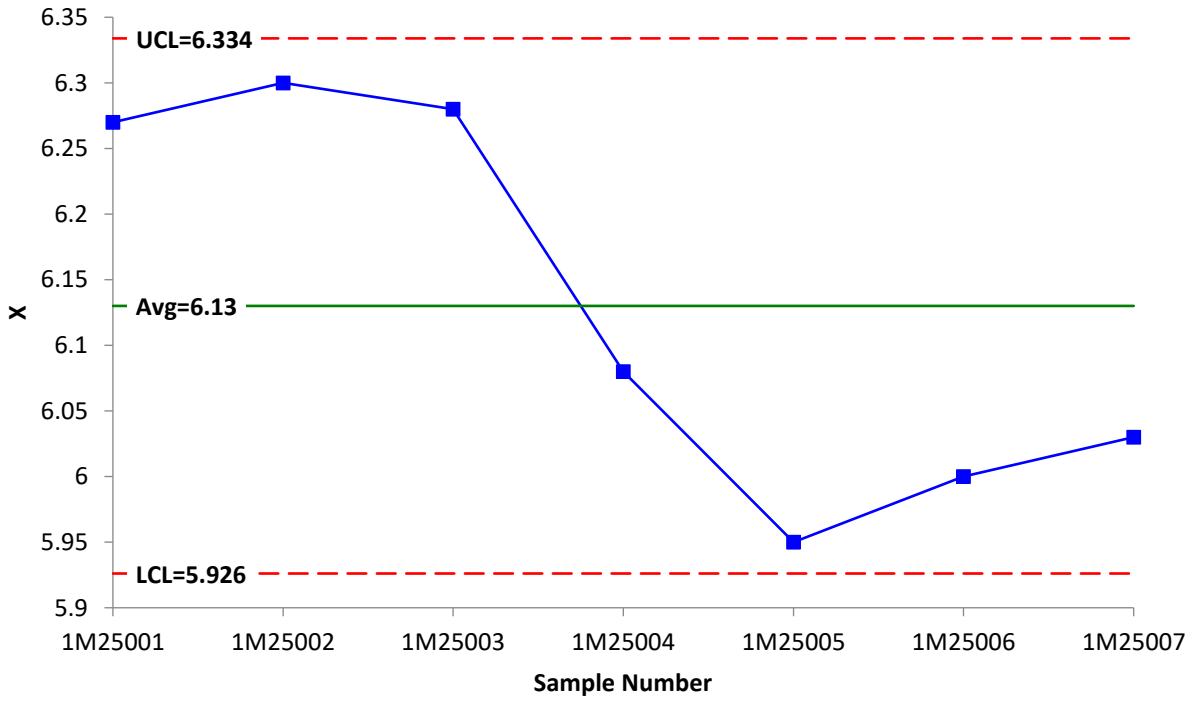
S/N	BATCH NO	Mannitol Assay (95 – 105%)	pH (4.5 – 7.5)	Sterility (MUST BE STERILE)	BACTERIAL ENDOTOXINS
1	1M25001	95.61	5.47	STERILE	<0.125
2	1M25002	95.53	5.57	STERILE	<0.125
3	1M25003	95.72	5.75	STERILE	<0.125
4	1M25004	96.44	5.72	STERILE	<0.125
5	1M25005	97.96	5.31	STERILE	<0.125
6	1M25006	99.8	5.15	STERILE	<0.125
7	1M25007	96.28	5.37	STERILE	<0.125

3.3.3 TREND ANALYSIS

Trend analysis of Mannitol content, pH, Pyrogen and Final yield %, were carried out on all the above batches.

Annexure I: pH trend for inprocess analysis of the product

Specification (4.5 -7.5**)**



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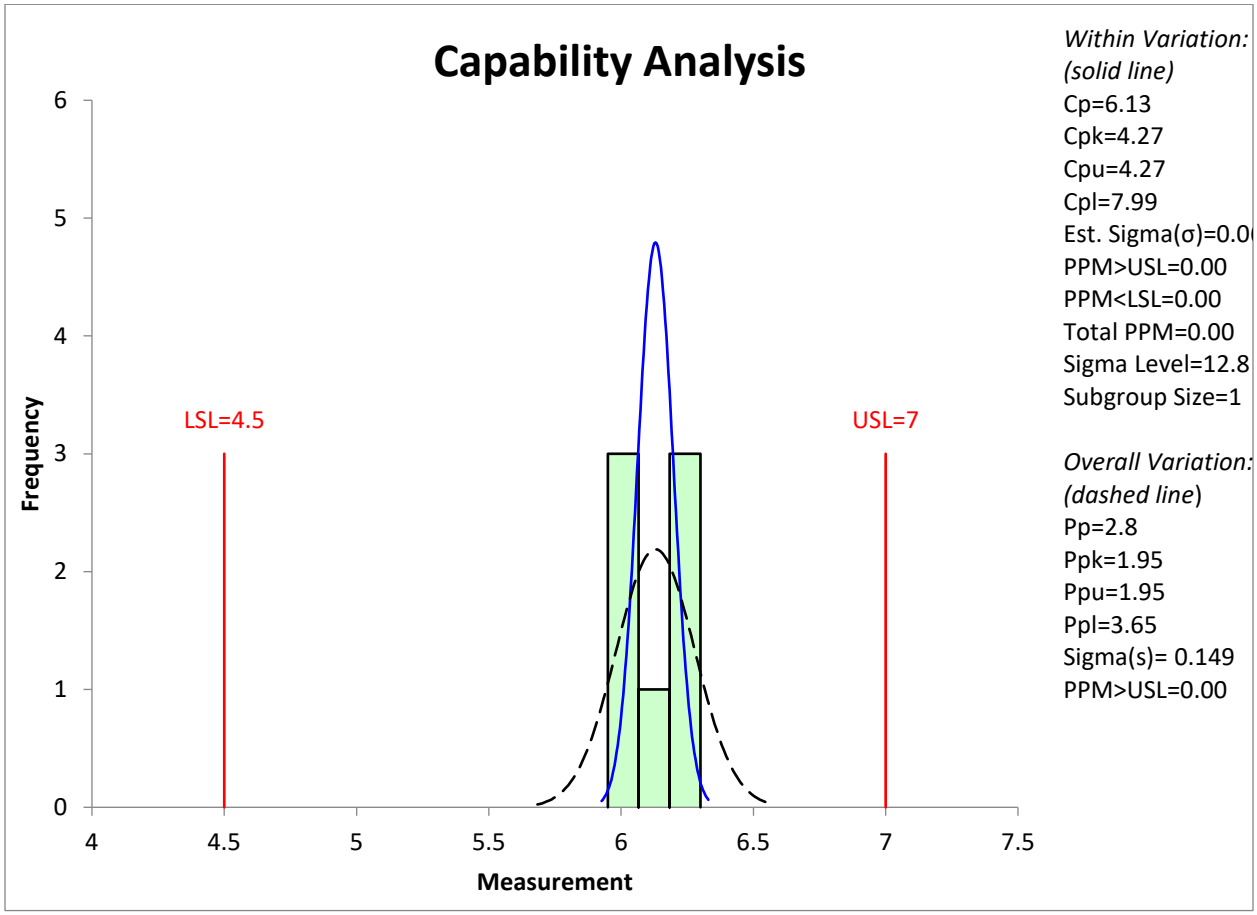

Mean	6.13
Standard deviation	0.148772757
Maximum	6.3
Minimum	5.95
Control limit: 5.68–6.58	
Alert limit: 5.83–6.43	
Minimum-Maximum: 5.95–6.3	
Alert limit minimum well within the specification limit, control limit existing within the specification limit.	

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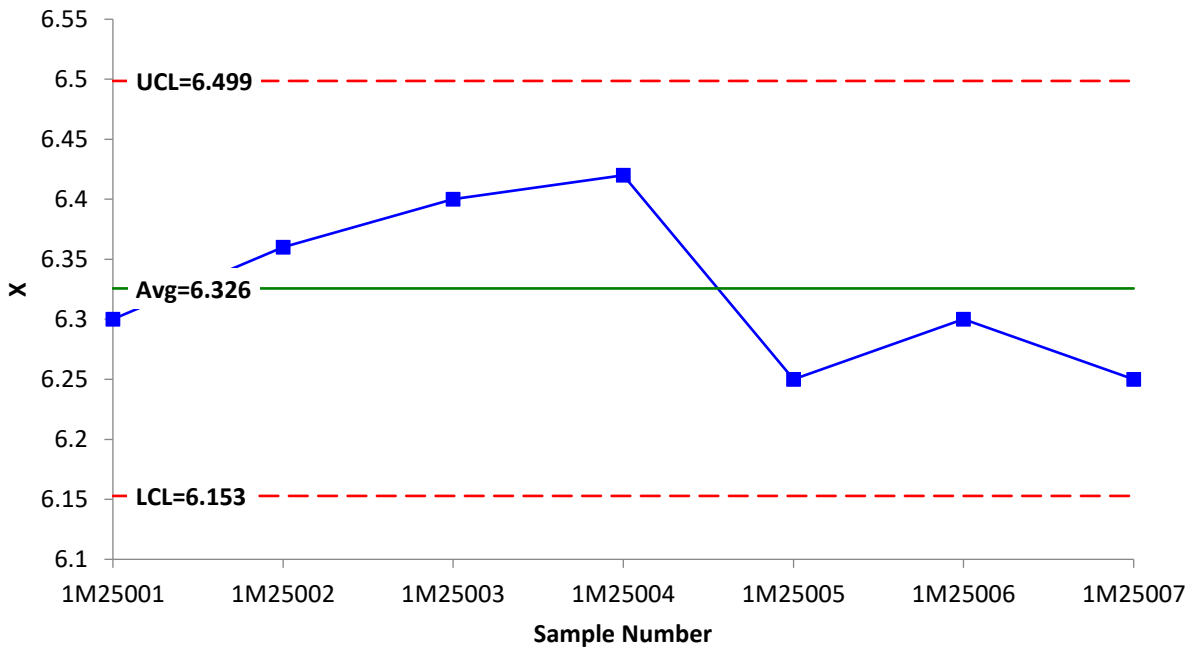
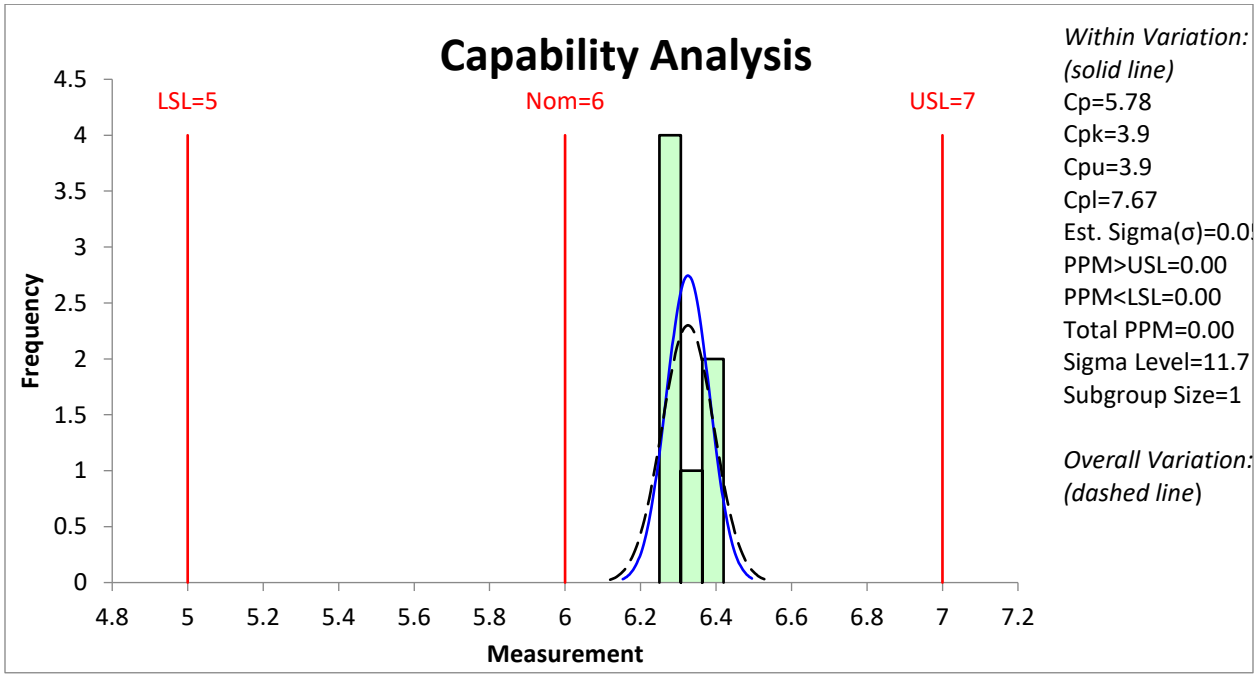
Annexure II: pH trend for Water for Injection (WFI)

Specification (5.0 -7.0)





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Mean	6.32571429
Standard deviation	0.06876461
Maximum	6.42
Minimum	6.25
Control limit: 6.12–6.53	
Alert limit: 6.19–6.46	
Minimum-Maximum: 6.25–6.42	
Alert limit minimum well within the specification limit, control limit existing within the specification limit.	

Annexure III: Assay trend of in-process analysis of Mannitol in Mannitol 10% Injection
Specification of Mannitol (95 – 105%**)**

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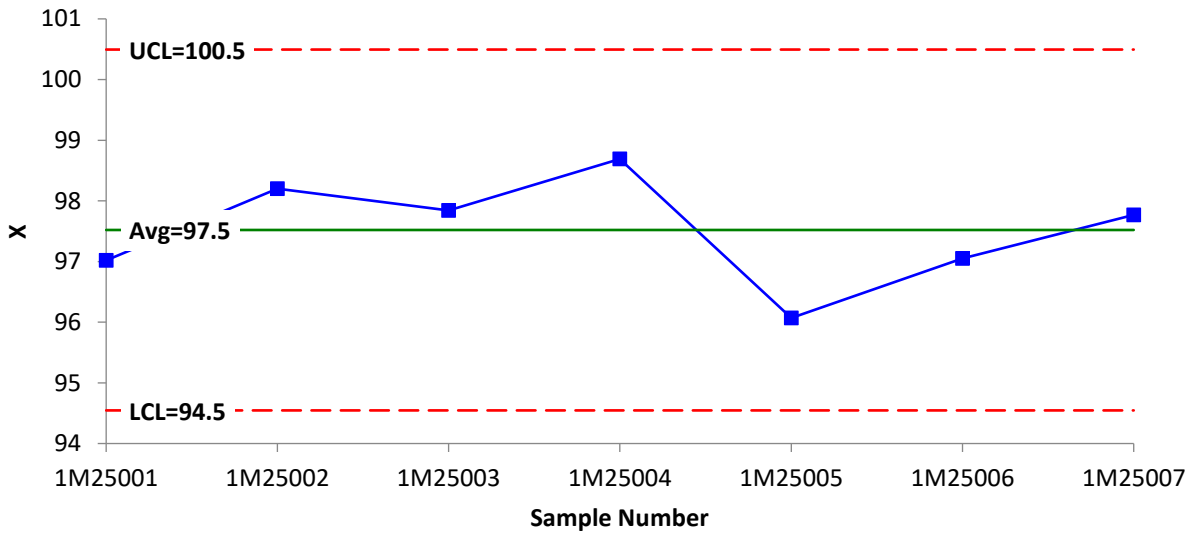

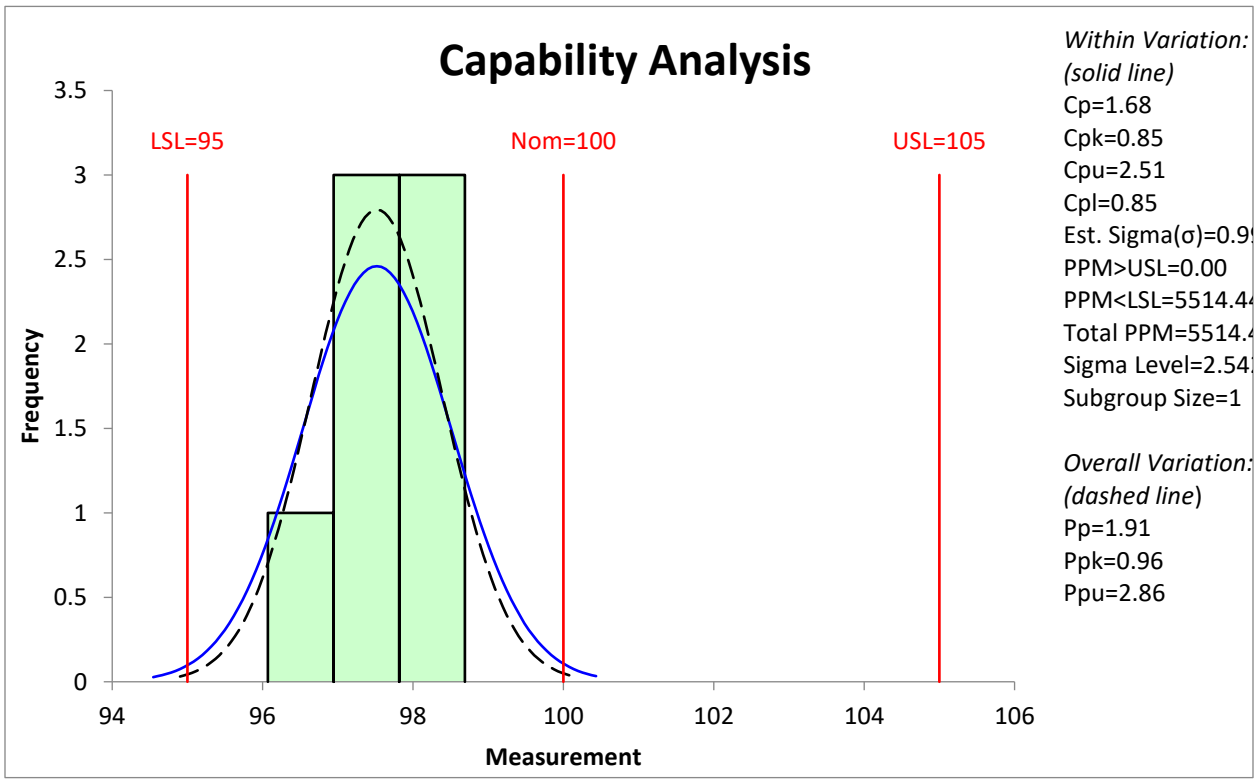

Mean	97.52
Standard deviation	0.872696969
Maximum	98.69
Minimum	96.07

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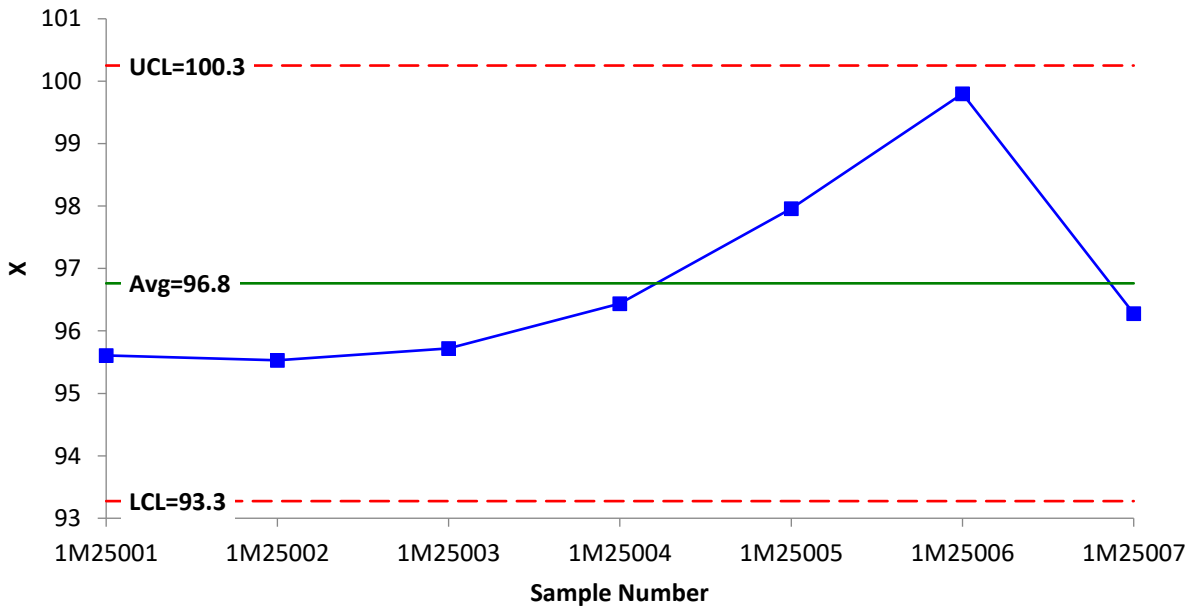
Control limit: 94.90–100.14
Alert limit: 95.77–99.27
'Minimum-Maximum: 96.07–98.69
Alert limit minimum within the specification limit, control limit existing along the specification limit.

Annexure IV: Assay trend of finished product analysis of Mannitol in Mannitol 10% Injection
Specification of Mannitol (95 – 105%**)**



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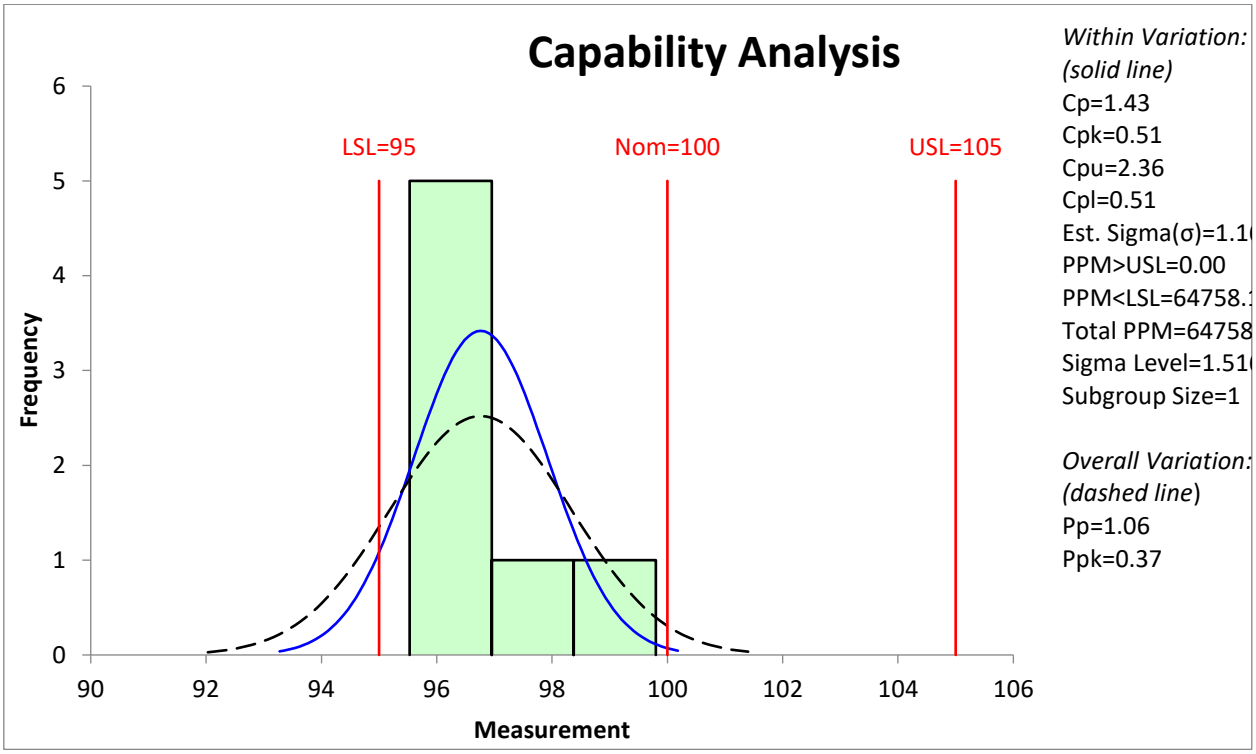

Mean	96.76285714
Standard deviation	1.577516131
Maximum	99.80
Minimum	95.53
Control limit: 92.03–101.50	
Alert limit: 93.61–99.92	
Minimum-Maximum: 95.53–99.80	
Alert limit minimum within the specification limit, control limit existing along the specification limit.	

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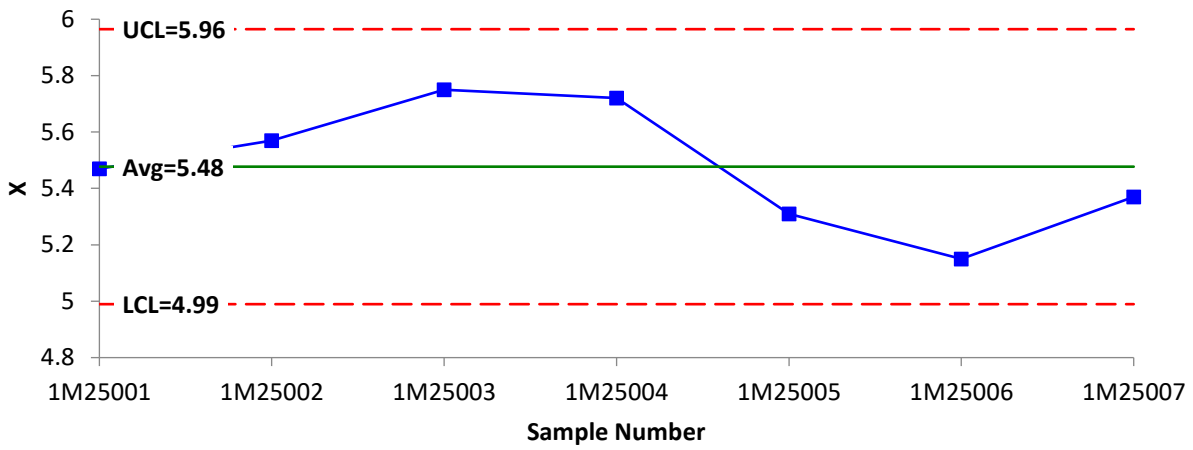
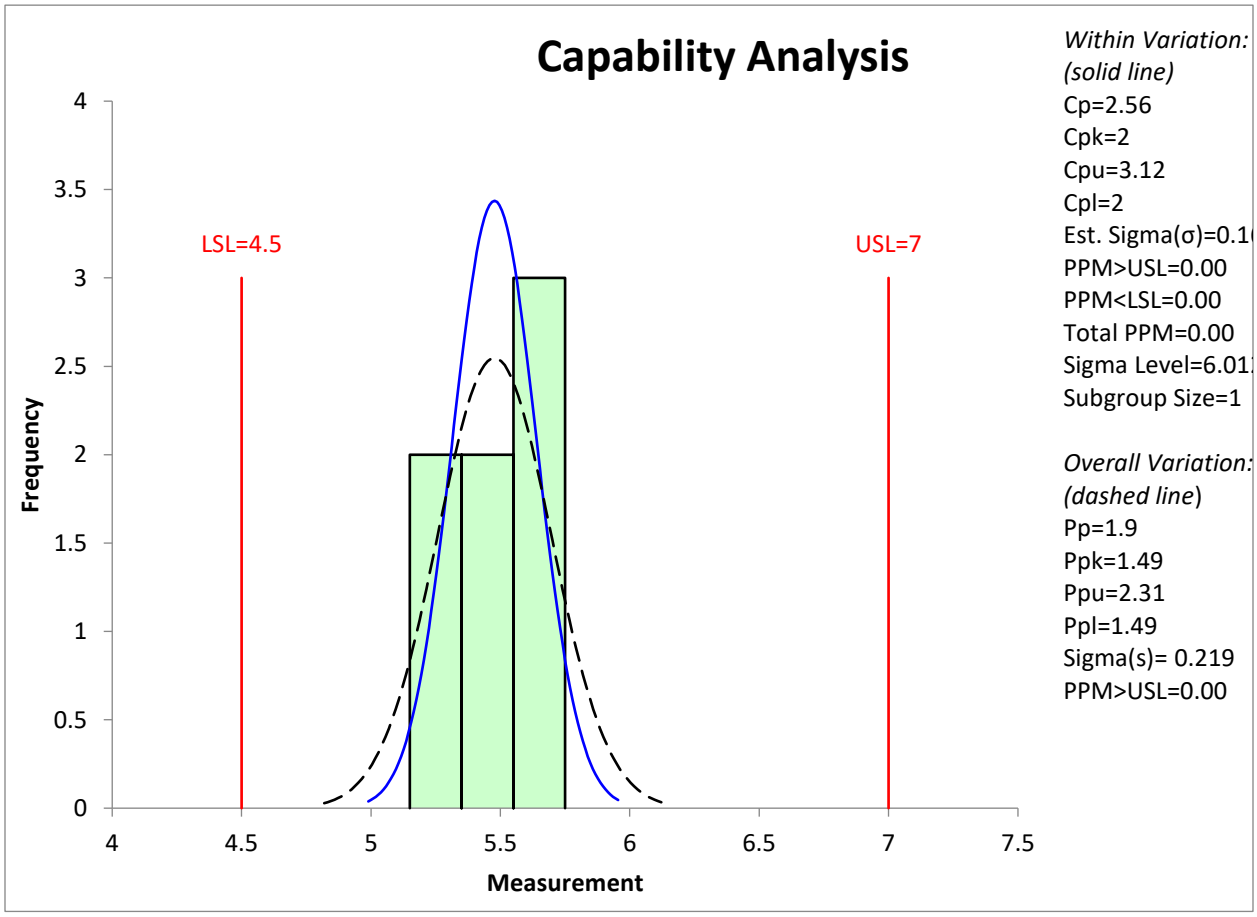
Annexure V: pH trend of finished product analysis of Mannitol 10% Injection

Specification of pH (4.50-7.50**)**





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Mean	5.477142857
Standard deviation	0.219295409
Maximum	5.75
Minimum	5.15
Control limit: 4.82–6.14	
Alert limit: 5.04–5.92	
Minimum-Maximum: 5.15–5.75	
Alert limit within the specification limit, control limit existing within the specification limit.	

SUMMARY OF PROCESS CAPABILITY

PROCESS CAPABILITY FOR INTERMEDIATE		
1	pH	4.27
2	Rinse Water	3.9
3	Mannitol	0.85
PROCESS CAPABILITY FOR PRODUCT		
1	pH	2.00
2	Mannitol	0.51

CQAs	Intermediate		Finished product	
	Control limit	Process capability	Process capability	Control limit
pH	CL within SL	4.27	2.00	CL within SL
Rinse water	CL within SL	3.9	-	-
Mannitol content	CL outside SL	0.85	0.51	CL outside SL

3.4 REVIEW OF BATCHES WITH OUT-OF-SPECIFICATION

No batch was recorded having out of specifications on any quality parameters.

3.5 REVIEW OF CRITICAL DEVIATIONS, NON-CONFORMANCES AND RELATED INVESTIGATIONS

3.5.1 PROCESS DEVIATIONS

No process deviation was reported from the standard processes for batches manufactured during the period Jan., 2025 to Dec., 2025.

3.5.2 YIELD DEVIATIONS

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No observed deviation from the predetermined tolerance was observed.

3.5.3 NON CONFORMANCE

No observed non-conformance from the predetermined tolerance was observed.

3.5.4 BATCH FAILURES

There was no batch failure observed during the period Jan., 2025 to Dec., 2025.

3.6 CHANGE CONTROL

There was no change done to the processes, equipment or analytical methods for the manufacture of Mannitol 10% Intravenous Injection during the year under review. As such, no change control was recorded throughout the period Jan., 2025 to Dec., 2025.

3.6.1 REGULATORY CHANGE

No regulatory changes as well as Master Formula Changes were made concerning these products throughout the manufactured batches till date.

3.6.2 CHANGES IN FORMULATION & PROCESS

No change is made in the formulation as well as process in regular practice. It has been observed that every batch manufactured, following consistent formulation and process, is giving consistent results which are proven by the trend analysis for all the parameters covered and expressed in graphical form.

3.7 STABILITY MONITORING PROGRAMME

The results of the real-time and accelerated stability monitoring program of all the commercial batches carried out showed consistency of all Critical Quality Attributes (CQAs). The product's Critical Quality Attributes (CQAs) like pH, Assay, Appearance, Pyrogenicity and Sterility are all found to be within the set specifications in conformity with the limits of the pharmacopeia. The stability report is attached for reference as BML/QA/SBP/001 (2025)

3.8 QUALITY-RELATED RETURNS, COMPLAINTS AND RECALLS

3.8.1 MARKET COMPLAINTS

Up till now there is no market complaint received for this product during the period Jan. 2025 to Dec. 2025.

3.8.2 RETURNED GOODS FROM MARKET

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No finished goods were returned from the market for the period Jan., 2025 to Dec., 2025 due to any quality problem

3.8.3 BATCH RECALLS

Up till now there is no batch recall for product during the period Jan., 2025 to Dec., 2025.

3.8.4 SALVAGED GOODS

There is no salvaging of goods for the period of Jan., 2025 to Dec., 2025.

3.9 CORRECTIVE ACTIONS

No critical, major or minor deviations have been observed during the year under review. As such, no noteworthy corrective action was necessary. Few observed deviations during the year under review fall under the category of Others, which has to do with colour or print on tertiary packaging.

3.10 VALIDATED ANALYTICAL AND MANUFACTURING PROCEDURES

MANUFACTURING PROCEDURE

Equipment cleaning and sterilization
Dispensing of the raw material
Solution preparation
Batch coding of PVC bag (packaging material)
Solution filling
Sterilization of the filled solution in qualified autoclave
Final packaging of the sterilized solutions

ANALYTICAL PROCEDURE

In-process testing of the cleanliness of equipment
In-process chemical testing of the semi-finished products for critical quality attributes like pH, assay
Chemical testing of the finished products
Microbiological tests on the finished product-
Sterility tests
Bacterial endotoxin tests (Pyrogen and Limulus Amoebocyte Lysate)

SUMMARY FOR PROCESS VALIDATION:

We have manufactured 8 batches of Mannitol 10% Intravenous Injection 500ml with same manufacturing formula since Jan., 2025 till Dec., 2025 and yearly APQR data shows satisfactory trend throughout the

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period. Also the retained samples prove that the product manufactured with the existing formulation is stable.

3.10.1.2 ENVIRONMENTAL CONDITIONS DURING MANUFACTURING OPERATION

The data of environmental conditions during the manufacturing of all batches were reviewed and recorded in the APQR. The environmental conditions of the critical areas were properly monitored in the raw material store, mixing unit, filling unit and batch coding unit

Environmental Conditions
Std Temp. : 25±2°C
Relative Humidity: 65%±5%

3.10.1.3 CRITICAL EQUIPMENT PERFORMANCE

The critical equipment used during the manufacturing and analysis of the product were identified and overall performance of these equipment were evaluated by reviewing the breakdown and preventative maintenance records.

3.11 REVIEW OF OBSERVED ADVERSE DRUG REACTIONS

No adverse drug reaction (ADR) was observed with the product during the year under review.

3.12 REVIEW OF PERIODIC RISK-BENEFIT EVALUATION REPORTS

3.13 REVIEW OF POST MARKETING SURVEILLANCE

The reports of clinical and therapeutic performance of the products from major health institutions across the nation have proven the product to be safe and efficacious. No adverse reaction has been reported, no report of product contamination nor pyrogenicity has been reported with the product within the year under review.

3.14 COMPARATIVE PRODUCT QUALITY REVIEW

CONCLUSION

Process capability analysis of the CQAs indicate, considering the number of batches produced for the year, variability in controls from the intermediate to the finished product stages. Heightened levels of control was witnessed for the Rinse water which translated into the control of the pH of the product through the stages. However, a more retrograding control was observed for the control of the mannitol content, exhibiting low capability at both the intermediate and finished stages. The increased variability may be influenced by one or more of dissolution, measurements, and weighting inconsistencies. The will require continued monitoring of this CQA to ensure a retrace to ensure improved capability.

Despite these observations, all batches produced within this period complied with the established specification for all tested parameters, and no out-of-specification was recorded.

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